S08. Gapped boundaries from exact anyon data

Pure Thought research portfolio

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Source: synthesis challenge 24; original synthesis. This topic is distinct from, or adds a distinct objective to, the canonical PRD catalog. Crosswalk

Plain-language guide

The problem in everyday terms

In some two-dimensional quantum phases, collective excitations behave neither like ordinary bosons nor fermions. Their possible combinations and exchanges restrict how the phase can end at an edge. This problem asks which proposed ways of absorbing these excitations at a boundary are mathematically consistent and can describe an edge with a nonzero energy gap.

Key terms

- **Topological order** — A type of quantum organization characterized by nonlocal structure and, here, anyon excitation data.
- **Anyon** — A two-dimensional excitation with exchange or braiding behavior beyond ordinary bosonic and fermionic possibilities.
- **Fusion** — The possible outcomes when excitations are combined.
- **Braiding** — The transformation associated with moving excitations around one another.
- **Topological twist** — The phase associated with a full rotation of an anyon.
- **F and R data** — Numbers or matrices specifying how fusion regrouping and exchanges act in chosen conventions.
- **Condensate** — A collection of excitations treated as absorbable into the boundary vacuum, together with consistent combination rules.
- **Lagrangian algebra** — The mathematical structure expressing a sufficiently complete compatible condensate for the boundary considered here.
- **Gapped boundary** — An edge whose excitations retain a positive energy threshold in the appropriate large-system limit.

Why this matters

Boundaries determine how a topological phase interfaces with its surroundings and which excitations can disappear or remain distinguishable there. Understanding these options supports the theory of topological phases and the organization of quantum information in such systems.

What progress would mean

Progress would verify a boundary candidate using complete consistency data, or reject it for a precise algebraic reason. Matching a list of fusion outcomes alone is insufficient. Turning the algebraic result into a particular microscopic boundary Hamiltonian is an additional construction.

Revised core question

For one fully specified two-dimensional topological order, can candidate condensates and their gapped boundaries be verified from exact categorical data?

Precise scope, target and outputs

Start with the toric-code modular category, explicit simple objects, fusion, twists, F and R symbols and conventions. Enumerate candidate condensable algebra objects within a declared multiplicity bound. Check multiplication/unit, associativity, commutativity and separability conditions, and the Lagrangian dimension condition for a boundary to vacuum. For the abelian toric-code benchmark recover electric and magnetic condensates. If constructing a lattice boundary, supply its Hamiltonian and check commuting-projector and gap claims separately.

Detailed critique of the synthesis framing

Fusion rules and modular S,T tables alone do not in general specify or prove all categorical coherence data. The synthesis also mixes 2D anyons with 3D topological orders, which require different structures. A listed condensate is not automatically a microscopic boundary realization. The rewrite retains an exact categorical task, bounds the enumeration and separates algebraic classification from lattice realization and anomaly/inflow interpretation.

Validation and milestones

Baseline: verify the toric-code coherence data and two standard vacuum boundaries. Strong: a second fully specified category or a completeness proof within a bounded class. Invalid F/R symbols and a nonbosonic condensate are rejection controls.

Use the shared evidence standard. Unsuccessful search is unresolved unless a complete exclusion proof is supplied. These are proposed baseline specifications, not completed scientific results.

Li, Yang and Dong, Gapped Boundaries of Kitaev's Quantum Double Models provides a primary example of linking Lagrangian algebras to explicit lattice boundary terms.

Displayed formulation and claim boundary

Supplemental specification.

$$A = \bigoplus_a n_a a, \qquad n_a \in \mathbb{Z}_{\geq 0}, \qquad (\dim A)^2 = \operatorname{Dim} \mathcal{C}.$$

For a vacuum boundary this Lagrangian dimension condition supplements, rather than replaces, the unit, multiplication, associativity, commutativity and separability checks using exact categorical coherence data. In the toric-code category the reference condensates are $1 \oplus e$ and $1 \oplus m$. A microscopic boundary Hamiltonian needs its own gap proof.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in

this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.